

## Earthquake Lite

Earthquake Lite is an App which displays global seismic activity, also displays the epicentre of the earthquake with the help of map view

## jStat

jStat tends towards building statistics and visualization software, making JavaScript and alternative for MATLAB or R

## Countermeasures to SQL injection

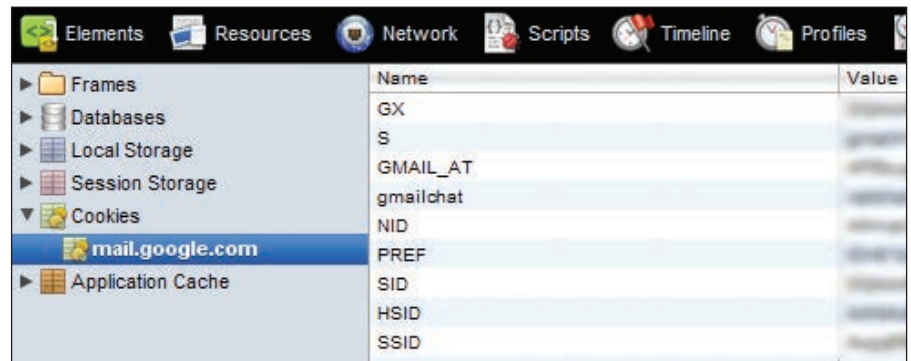
There are a few points to be mindful of. The first is that for the above trick to work, you must know the table column names that are used to store the usernames and passwords, which is not too tough to guess because most database designers would choose obvious and easy to guess names.

There is very little a user can do, but if you are a developer, we have something to say. First, the trick fundamentally works depending on what DBMS you might be bringing into use. Not all databases read the AND and OR keywords in same style. Depending on the DBMS used, you might like to amend the position of these keywords. Second, different databases treat the newlines within the query differently. While some databases might recreate the complete query as a single liner, others would hate to do this ever. So please read the manual. As a third point, the trick basically works because you have been able to guess the table's column names. Changing them to something which cannot be easily guessed, you might make the work overly complicated for the attacker. For example, if you change the column names to something like user-name\_3397 and passwords\_9865, they do not remain easy to be guessed. Do they?

## Session Hijacking

As the term suggests, it depends on 'sessions'. Whenever you log into a website, the website creates a new session for you. Most websites create a cookie (don't get mistaken with the yumminess) on your computer which is used only for that session. When you log out, your session ends and cookie expires. Sometimes people are almost always logged into some website. An expired cookie is useless; but it is quite important if you are still logged in.

Take your web mail for example. You do not want to log in again and again so you check that 'keep me signed in' or 'Remember me' checkbox and login. Congrats, a cookie has been embodied on your computer which would not expire for a long time. This cookie can now be used on another machine to gain access



Some details you overlook...

to your account, without knowing your password (and that is the scariest part). If a malicious attacker gains access to your account just once, he can change your password or download everything from your account on his computer and you will have no idea about it. Now that's what you call espionage.

It is not a big deal to steal cookies from your computer. This job is an easy meat for a moderate level hacker. Not only this, they can be seen with naked eyes using any modern web browser. For example, in Google Chrome, right click anywhere and select "Inspect Element". This opens developer tools on the lower half of the window. Now click on the resources section on the top bar of the developer tools section. It will list the different types of resources that Google Chrome manages for you. On the left pane, you will find an expandable menu for 'Cookies' where you can see the cookies set by the website itself as well as by other sites whose elements are displayed on the page (e.g. Facebook Social Plugin). If you are wondering about Firefox, here are the links to help you (it's called SQLite Manager):

<https://addons.mozilla.org/en-US/firefox/addon/5817>

[http://kb.mozillazine.org/Profile\\_folder\\_-\\_Firefox](http://kb.mozillazine.org/Profile_folder_-_Firefox)

We believe that any reader aware of the web security 'like you' can make out the rest by himself.

Not every cookie which is enlisted there is important but someone, who has spent some time can easily figure out which one is important and gobble it easily. Methods to steal a cookie can vary, but the effect will be same; your

account being compromised, hacked down, and used by someone else. Although most major sites have their defences up against this technique, some are unaware, some are too indifferent.

There is yet another type of session hijacking done via URLs. At many occasions, you might have seen something like `sessid=1ca73292fae72da6ec02c1f17bb61902` on your URL bar. Actually that 'crap' that you are watching is the session ID generated by the website for you. Some (or should we say 'most') users share URL on the web without noticing this part. Feeling déjà-vu? Although most websites using a popular CMS software are safe from the attack, every once in a while some hacker with some innovative material inside his skull comes up with a method to crack the nut. If such a URL is shared and the programs working on the website use only a session ID to recognize and differentiate a user from another, the user to whom the session ID actually belongs is risking his privacy to a great extent.

## Countermeasures

Session Hijacking concerns cookies and URLs. As far as URLs are concerned, all you have to take care is to not share URLs which contain your session ID on the web, not at least publicly. Sometimes just removing the part of the URL containing your session ID can work. As far as cookies are concerned, the best way around is to make sure that people other than you are not able to access them. Although paranoid, the best solution is to encrypt the partition in which the cookies rest. With professional and ultimate editions of Win-